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Automation of Standard Technological Processes and Manufacturing

**DEVELOPMENT OF AN AUTOMATED PRESSURE CONTROL SYSTEM
FOR URBAN WATER SUPPLY PUMPING STATION**

SIS № 1,2

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INTRODUCTION

In modern industrial engineering, the automation of technological processes plays a crucial role in enhancing operational efficiency, reducing energy consumption, and ensuring the reliability of continuous production systems. Water supply networks represent a primary class of critical municipal and industrial infrastructure where dynamic load variations require robust automated control strategies. Uncontrolled fluctuations in water consumption lead to excessive energy expenditure, severe hydraulic shocks, and accelerated physical degradation of mechanical components.

To address these challenges, advanced automation frameworks rely on frequency-controlled electric drives integrated with feedback control loops. The objective of this study is to perform a comprehensive technological and mathematical analysis of a water supply pumping station equipped with an adjustable frequency converter (FC) and induction motor drive.

This research synthesizes the domain knowledge from recent peer-reviewed scientific literature to establish a rigorous system model. By representing the pumping station as an input-output "black box" control plant, this work formalizes the complex multi-stage dynamics of the electro-hydraulic system. Specific focus is placed on mapping the interactions between manipulated control inputs, non-stationary environmental disturbances, and internal system variables. The ultimate purpose of this foundational structure is to lay the theoretical groundwork for closed-loop control system design, parameter optimization, and energy-efficient pressure stabilization under stochastic daily demand.

1 DESCRIPTION OF THE TECHNOLOGICAL PROCESS

This chapter describes the technological process of the automated urban water supply pumping station, including the mechanical equipment, filtration system, storage reservoir, and pressure control mechanisms. It also presents the structural arrangement, process block diagram, and key operational parameters required for the control system design.

1.1 Description of the Urban Water Supply Process

The technological process begins with the raw water intake from the primary storage supply (RESERVOIR). Water passes through an automatic mechanical filter unit (FILTER) to remove suspended particulate impurities, protecting downstream hydraulic components and preventing sediment accumulation within the pump impellers.

The pumping station utilizes a parallel configuration of centrifugal pumps (P-1 and P-2) driven by three-phase induction motors (M-1 and M-2). Under nominal operation, each pump is designed for high-efficiency hydraulic delivery, with one pump running as the primary operating unit while the second remains in standby or active staging mode to guarantee system redundancy and satisfy peak volumetric demand. To isolate individual lines during maintenance and prevent backflow during emergency pump stops, manual isolation valves (BV-1 to BV-4) and check valves (CV-1, CV-2) are integrated into the pipeline manifold. A dedicated drain line (DRAIN) with isolation valve BV-4 allows full system flushing.

Each pump motor is driven by an individual Variable Frequency Drive (VFD-1 and VFD-2). The VFDs regulate the rotational velocity of the pump shafts by adjusting the output power supply frequency ($f_{VFD} = 0 - 50 \text{ Hz}$), enabling precise fluid pressure modulation and minimizing dynamic pressure surges (water hammer).

The primary controlled process parameter is the water pressure in the main discharge header (*P_{actual}*) supplying the urban distribution network (URBAN WATER SUPPLY). A suction pressure transmitter (PT-1) monitors inlet pressure from the reservoir to protect against cavitation, while a discharge pressure transmitter (PT-2) continuously measures actual network delivery pressure (*P_{actual}*) and provides analog feedback (4–20 mA) to a Programmable Logic Controller (PLC). Additionally, an electromagnetic flow transmitter (FT-1) captures instantaneous volumetric delivery flow rate.

The measured output pressure from PT-2 is compared against the operator-defined pressure setpoint ($P_{set} = 4.5 \text{ bar}$) within the PLC. The closed-loop control algorithm continuously calculates the dynamic pressure deviation error:

$$\Delta P = P_{set} - P_{actual}$$

If consumer demand increases and *P_{actual}* drops below P_{set} , the controller increases the VFD output frequency, accelerating motor speed to restore target network

pressure. Conversely, when network consumption decreases and P_{actual} exceeds P_{set} , the controller decreases frequency to conserve electrical energy.

To ensure safe operation, emergency protection interlocks are integrated into the control routine. If line pressure violates allowable limits ($P_{actual} < P_{min}$ or $P_{actual} > P_{max}$), or in cases of motor current overload or dry running, safety interlocks immediately halt the pump drives, isolate emergency valves, and activate audible and visual alarm signals (ALARM). The core feedback loop sequence is formalized as:

PT – 2 → *PLC* → *VFD* → *Induction Motor (M)* →
Centrifugal Pump (P) →
→ *Discharge Pressure P_{actual}*

1.2 Process & Instrumentation Diagram (P&ID) Architecture

To synthesize the operational interaction between the hardware components, electrical power distribution, hydraulic piping, sensing instrumentation, and automation logic, a comprehensive structural architecture is established. This multi-tier framework integrates field hardware, power actuation, sensing instrumentation, and supervisory control into a unified automated system. The intake hydraulic line connects the primary storage reservoir to the pumping manifold through an automated filtration system designed to eliminate particulate matter and protect downstream impellers.

Fluid flow is propelled by two centrifugal pumps driven by three-phase asynchronous motors, while mechanical isolation for maintenance is provided by ball valves, and non-return check valves prevent hazardous backflow through inactive pump lines. Main electrical power feeds supply two independent Variable Frequency Drives through circuit protection breakers, enabling continuous modulation of output frequency to dynamically adjust motor shaft speeds, suppress mechanical stress, and eliminate water hammer transients during pump activation. Continuous field monitoring is maintained by analog transmitters, where a suction pressure transmitter monitors intake head pressure to protect against cavitation, a discharge pressure transmitter measures network delivery pressure, and an inline electromagnetic flow transmitter measures volumetric discharge rate.

All field sensor signals are routed via standardized current loops to the central Programmable Logic Controller, which executes closed-loop control routines and computes real-time frequency setpoints sent directly to the drives. At the supervisory tier, a Human-Machine Interface panel and a centralized SCADA workstation communicate with the controller via industrial communication buses for real-time process visualization, trend archiving, and operator setpoint adjustments. In the event of system faults or operational limit violations, the controller triggers visual and audible alarm beacons and executes automated emergency interlocks. The complete structural and functional topology integrating all hardware equipment, sensors, actuators, power distribution lines, and supervisory control buses is graphically summarized in **Figure 1.1** below.

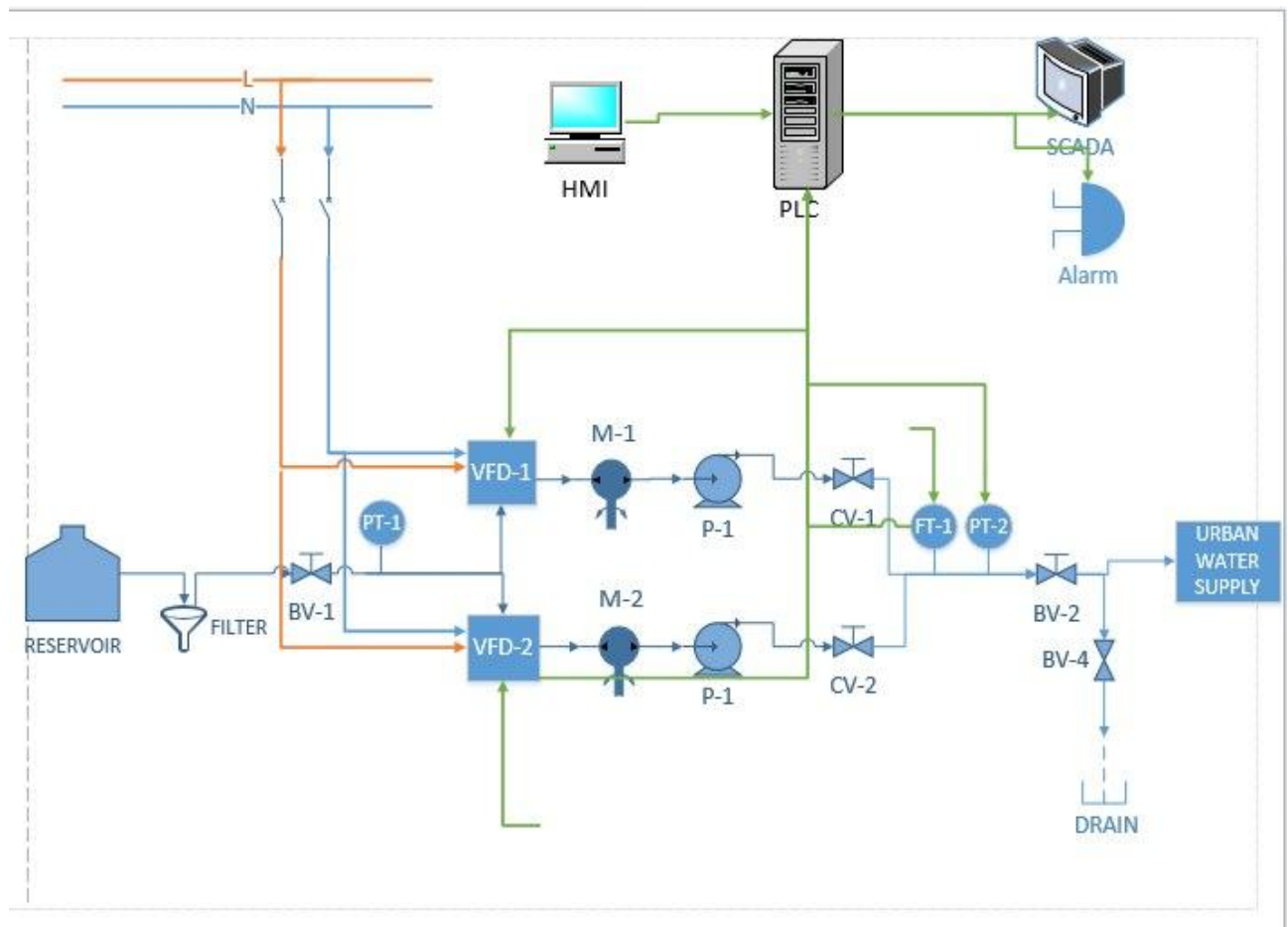


Figure 1.1 – Complete Process & Instrumentation Diagram (P&ID) of the Automated Pumping Station

1.3 Algorithmic Flowchart and Control Logic Analysis

To establish a fully automated, safe, and energy-efficient operation, the physical equipment of the water supply pumping station is integrated with a multi-tier instrumentation, control, and supervisory system. The functional and operational interactions between electrical power lines, fluid conduits, field transmitters, programmable controllers, and supervisory operator stations are structured into three distinct operational levels:

1. Instrumentation and Actuator Level (Field Level):

Sensing Infrastructure: The intake line is equipped with a suction pressure transmitter (PT-1) to measure reservoir outlet pressure and protect the pumps against low-suction conditions. The main distribution header features a discharge pressure transmitter (PT-2) for continuous process feedback, as well as an electromagnetic flow transmitter (FT-1) for measuring discharge volume.

Actuation Infrastructure: Power lines (L/N) feed two frequency converters (VFD-1 and VFD-2), which supply variable frequency voltage to induction motors M-1 and M-2. Mechanical isolation is provided by ball valves (BV-1 to BV-4), while check valves (CV-1 and CV-2) prevent backflow through inactive pump channels.

The executive field-level feedback loop, which directly processes real-time sensor measurements (PT-1, PT-2, FT-1) and adjusts VFD frequencies to maintain Pset, is illustrated in **Figure 1.2** below.

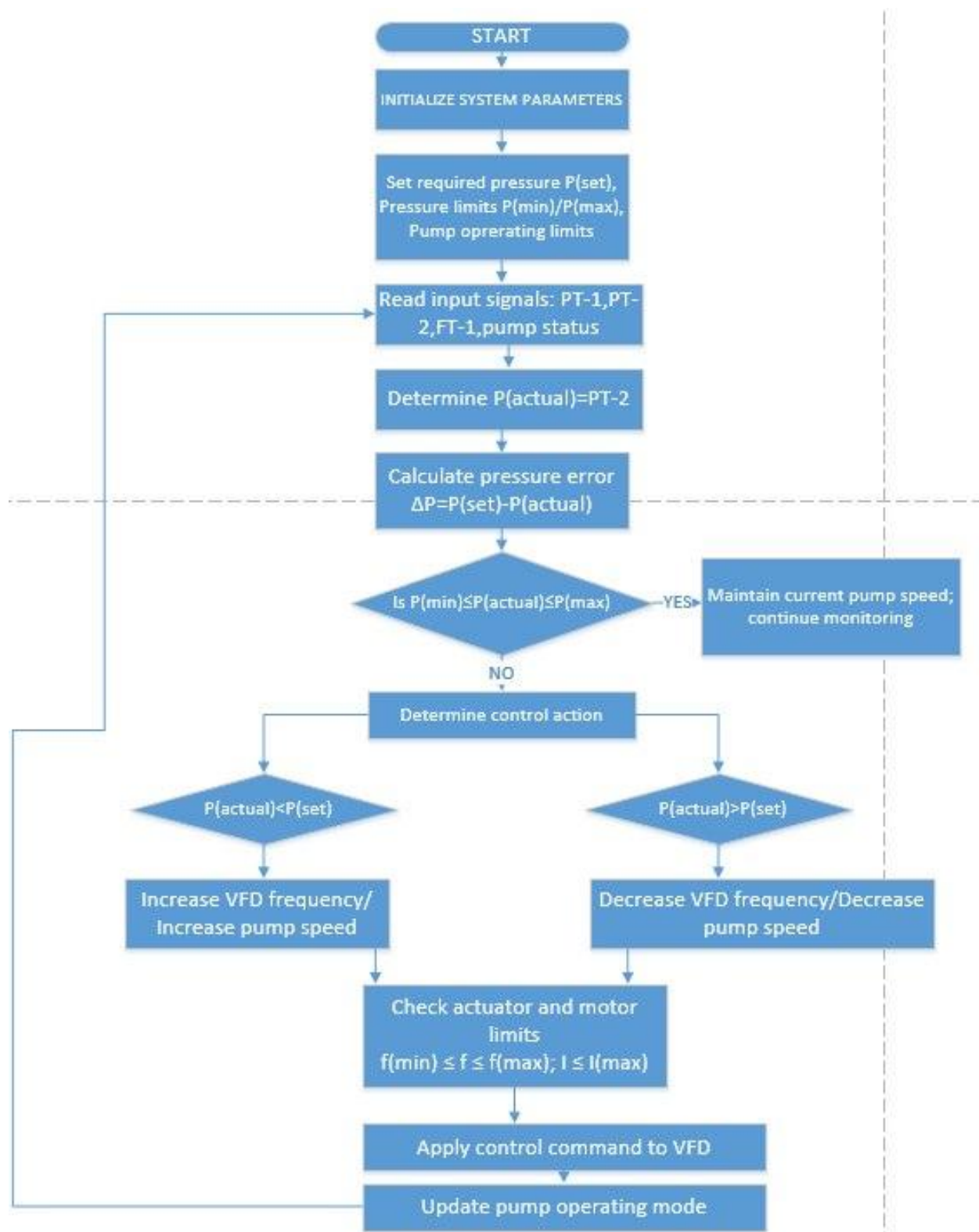


Figure 1.2 – Field-Level Control Loop for Pressure Regulation and VFD Command Execution

2. Control Level (Automation Level):

The core of the control layer is a central Programmable Logic Controller (PLC). The PLC receives continuous analog feedback (4–20 mA) from field sensors (PT-1, PT-2, FT-1) and status signals from the drives.

It executes the closed-loop Proportional-Integral-Derivative (PID) control logic, calculates frequency command signals, filters measurement signals, and performs safety diagnostics. In case of emergency conditions (overpressure, dry running, motor overload), protective interlocks immediately halt the drives and close emergency valves.

The logical execution flowchart of signal filtering, operating indicator calculations, emergency interlocks, and database logging at the PLC level is presented in **Figure 1.3** below.

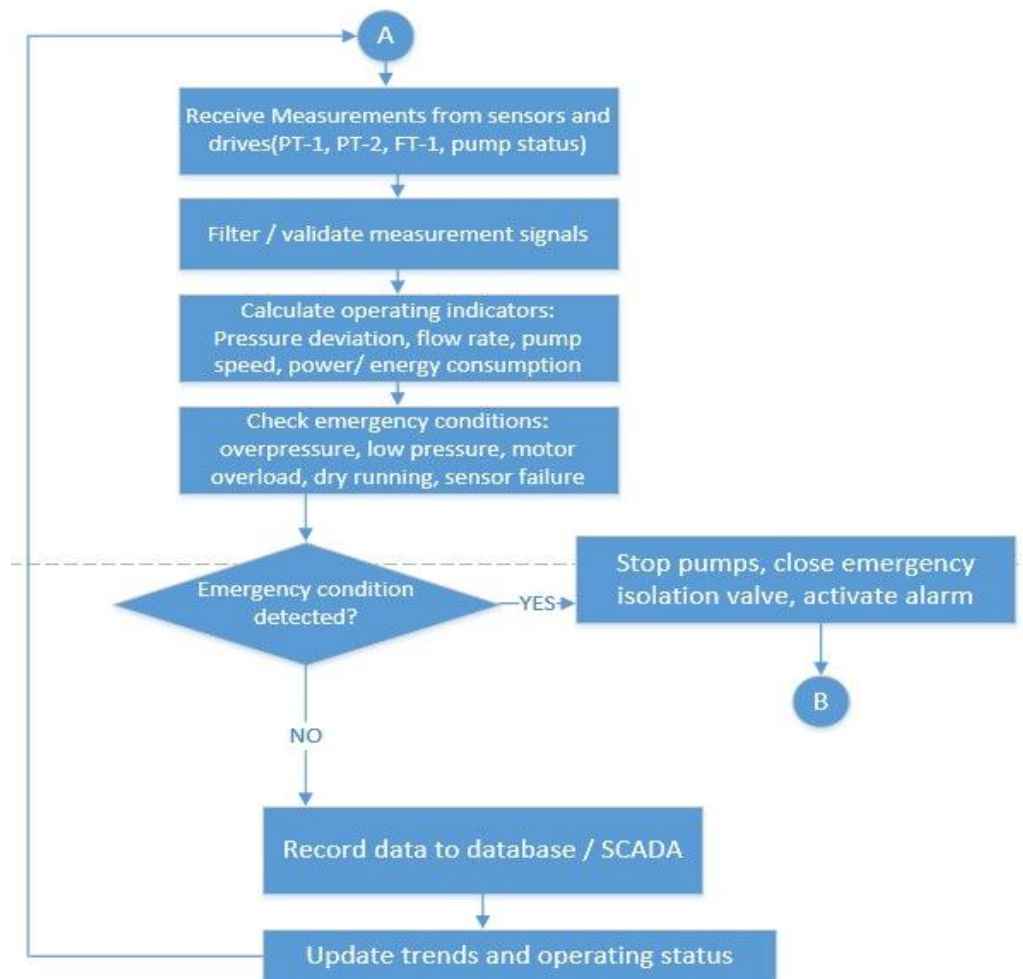


Figure 1.3 – Control-Level Flowchart for Signal Diagnostics, Protective Interlocks, and Alarm Handling

3. Supervisory and Management Level (SCADA/HMI Level):

The top layer consists of a Human-Machine Interface (HMI) screen and a centralized SCADA system workstation connected to the PLC.

This interface provides real-time process visualization, trends monitoring, historical data logging, and setpoint modification (Pset). In the event of system faults, an audible and visual alarm beacon (Alarm) is triggered by the controller to alert dispatchers.

The supervisory interactions, operator setpoint adjustments, historical archiving, and reporting algorithms executed at the SCADA level are illustrated in **Figure 1.4** below.

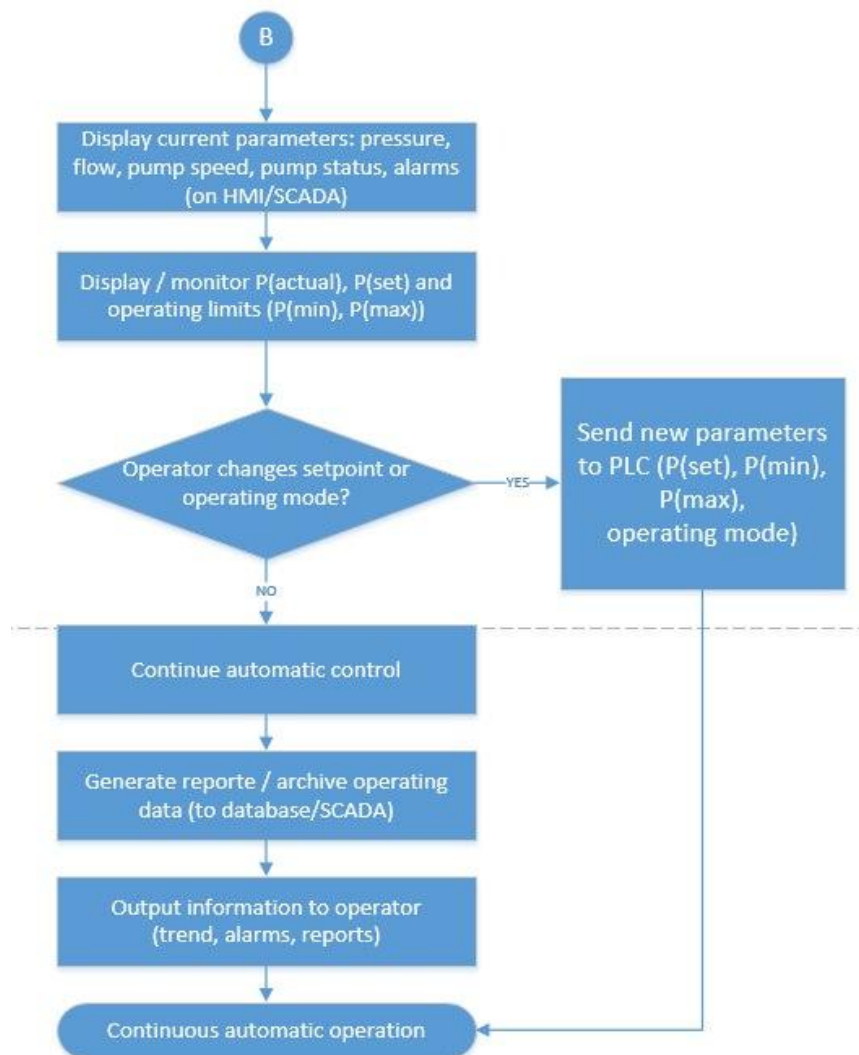


Figure 1.4 – Supervisory-Level Flowchart for HMI Visualization, Setpoint Management, and Archiving

1.4 Review of Modern Control Approaches for Water Stations.

The design of automated water supply control systems has evolved from traditional open-loop manual switching to advanced closed-loop feedback algorithms. In modern municipal water distribution stations, maintaining constant head pressure under highly non-stationary and stochastic consumer demand requires robust, high-precision control strategies.

1. **Conventional PID Control (Proportional-Integral-Derivative):** PID control remains the industry standard in Programmable Logic Controllers (PLCs) due to its structural simplicity, high reliability, and low computational overhead. In Variable Frequency Drive (VFD) pumping systems, PID controllers continuously calculate the error between target setpoint (P_{set}) and actual line pressure (P_{actual}). However, fixed-parameter PID controllers struggle with non-linear hydraulic dynamics, transport delays, and severe pipe network pressure surges (water hammer).
2. **Variable Frequency Drive (VFD) Integration:** Integrating frequency converters with induction motors allows continuous modulation of pump shaft rotational speed ($f=0-50$ Hz). Compared to traditional throttling valve regulation, VFD-based control reduces active electrical power consumption by up to 30–40% according to affinity laws, eliminates mechanical start-up stress, and significantly suppresses hydraulic shock waves.
3. **Advanced and Intelligent Control Strategies:** To overcome the limitations of classical PID algorithms in complex hydraulic networks, modern control engineering utilizes adaptive and predictive techniques:
 - **Model Predictive Control (MPC):** Uses explicit dynamic models of the pipeline network to predict future pressure behavior and optimize VFD control outputs over a moving time horizon.
 - **Fuzzy Logic and Neural Network Control:** Enables adaptive setpoint regulation (P_{set}) based on real-time flow measurements (FT-1) without requiring explicit dynamic differential equations, optimizing energy consumption during low-demand night hours.

In this work, a classic discretized PID regulation algorithm integrated with emergency protective interlocks and SCADA setpoint supervisory management is adopted as the primary control architecture.

2 FORMALIZATION OF THE CONTROL OBJECT

This chapter establishes a mathematical and structural formalization of the automated water supply pumping station as a dynamic control object. The electro-hydraulic plant is mapped using an input-output "black box" representation, boundary conditions are defined, and system variables are classified to enable controller synthesis.

2.1 "Black Box" Model and Boundary Parameters Definition

The pumping station is a complex, non-linear electro-hydraulic plant characterized by multi-domain physical interactions, including electromagnetic conversion in induction motors, centrifugal fluid dynamics, and non-stationary pipe hydraulics. Standard control system design requires translating these multi-physics phenomena into a structured mathematical framework. However, deriving exact analytical equations for continuous fluid mechanics and wave propagation in pipeline networks often leads to excessively high-order dynamic models that are impractical for real-time Programmable Logic Controller (PLC) execution.

To synthesize an effective dynamic control system without overcomplicating the internal physical dynamics of hydrodynamics and electromagnetism, the pumping station is formalized as a dynamic "Black Box" system. In this model, the system is viewed through its boundary interactions, mapping direct relationships between control inputs, external disturbances, internal state parameters, and output variables, as illustrated in **Figure 2.1** below.

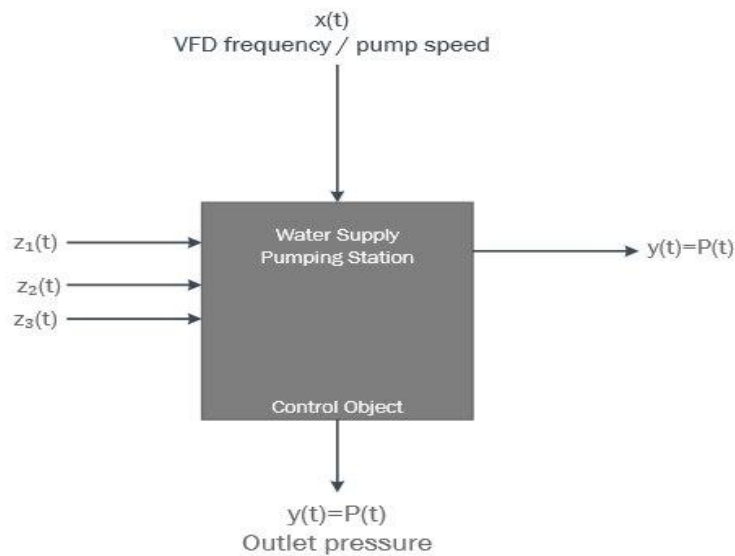


Figure 2.1 – Input-Output "Black Box" Representation of the Pumping Station

2.2 Specification and Classification of Control System Variables

Based on the formalized "Black Box" model of the automated pumping station, the system operational parameters are classified into input control signals, output controlled variables, setpoints, measured parameters, and external disturbances. This functional breakdown is summarized in **Table 2.1** below.

Parameter	Symbol	Description	Role in the system	Signal Type (I/O)
Control input	$x(t)$	VFD frequency / pump rotational speed	Changes the operating mode of the centrifugal pump	Output, analog
Controlled output	$y(t)$	Outlet water pressure	Main controlled parameter of the pumping station	Input, analog
Disturbance	$z(t)$	Variation of water demand	Changes the required pump performance	Unmeasured, dynamic
Disturbance	$z(t)$	Variation of inlet pressure	Influences the operating condition of the pumping station	Input, analog
Disturbance	$z(t)$	Hydraulic changes in the pipeline network	Changes pressure and flow conditions in the system	Unmeasured, dynamic
Setpoint	Pset	Required outlet water pressure	Reference value for the automatic control system	Internal, analog
Measured value	$P(t)$	Actual outlet water pressure measured by pressure sensor	Feedback signal for the controller	Input, analog

Table 2.1 – Input, Output and Disturbance Parameters of the Control Object

Detailed Functional Description of Variables

1. **Control Input ($x(t)$):** The primary manipulated variable generated by the controller and applied to the Variable Frequency Drive (VFD). By adjusting the output frequency $x(t)$, the rotational speed of the pump impeller changes, directly modulating the hydraulic energy supplied to the fluid flow.
2. **Controlled Output ($y(t)$):** The continuous dynamic parameter of the process that defines system performance. The target of the control system is to stabilize the outlet pressure $y(t)$ at a constant value under dynamic operating conditions.
3. **External Disturbances ($z(t)$):** Uncontrolled environmental and consumer load fluctuations acting on the pumping station:
JPEG
 - Water consumption dynamics ($z1(t)$) causing immediate header pressure drops.
 - Fluctuations in intake reservoir pressure ($z2(t)$) affect the overall head generated by the station.
 - Dynamic structural changes in the pipeline distribution topology ($z3(t)$) altering flow friction and hydraulic resistance.
4. **Setpoint (P_{set}) and Measured Value ($P(t)$):** The setpoint P_{set} represents the target outlet pressure defined by operational requirements. The pressure transmitter PT-2 continuously measures actual line pressure $P(t)$ to provide a feedback signal to the Programmable Logic Controller (PLC). The active error signal processed by the control algorithm is defined as:

$$\Delta P(t) = P_{set} - P(t)$$

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